LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	23
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Gyroscope as an attitude-sensing device, establishment of attitude reference by displacement gyro and rate gyro.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Gyroscope as an attitude-sensing device, establishment of attitude reference by displacement gyro and rate gyro.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagrams and gyro sensor locations in IB; Instructor supervision during Q&A.;
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a gyroscope in an aircraft?
(b) How does the conservation of angular momentum apply to flight instruments?
(c) Why is an attitude reference necessary for the AFCS?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic Gyroscopic Properties • Rigidity in Space • Precession Principles • Gyroscopic Torque • Construction of Displacement Gyro	L & Ds	PPT, IB	15 min
(b)	• Displacement Gyro Operation • Attitude Reference Establishment • Rate Gyro Functionality • Rate Gyro Construction • Damping Mechanisms	L & Ds	PPT, PDF	20 min
(c)	• AFCS Integration of Gyros • Combined Displacement and Rate Use • Error Sources in Gyro Systems • Gyro Alignment Procedures • System Feedback Loops	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a displacement gyro establishes an attitude reference.
(b) Describe the role of a rate gyro in the AFCS system.
(c) How do displacement and rate gyros work together in modern AFCS?

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of displacement and rate sensing in this lesson will be applied in the next lesson regarding single axis and combined rate sensing.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Rate sensing about three axes, single axis rate sensing, combined use of rate and displacement gyros.	Refer to the master syllabus reference list	LP 24

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min